

SESSION – 10

Java Dequeue

```
import java.io.*;
import java.util.*;

public class Solution {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int m = sc.nextInt();

        Deque<Integer> deque = new ArrayDeque<>();
        HashMap<Integer, Integer> frequency = new HashMap<>();

        int maxUnique = 0;

        for (int i = 0; i < n; i++) {

            int value = sc.nextInt();

            // Add new element
            deque.addLast(value);

            // Update frequency
            frequency.put(value, frequency.getDefault(value, 0)
+ 1);

            // When window becomes larger than m
            if (deque.size() > m) {

                int removed = deque.removeFirst();

                int count = frequency.get(removed);

                if (count == 1) {
                    frequency.remove(removed);
                } else {
                    frequency.put(removed, count - 1);
                }
            }

            // Number of unique elements in current window
```

```

        if (deque.size() == m) {
            maxUnique = Math.max(maxUnique,
frequency.size());
        }
    }

    System.out.println(maxUnique);

    sc.close();
}
}

```

Congratulations

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Test case 0

Test case 1 [🔒](#)

Test case 2 [🔒](#)

Test case 3 [🔒](#)

Test case 4 [🔒](#)

Test case 5 [🔒](#)

Test case 6 [🔒](#)

Compiler Message

Success

Input (stdin)

```

1 6 3
2 5 3 5 2 3 2

```

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Expected Output

```

1 3

```

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